

Homework 3

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Table of contents

Set-Up Tasks	2
Problem 1	2
a. Appropriate test	2
b. Visualization	2
c. Assumption checks	3
Check assumptions	3
Run test	5
d. Communication results	6
e. How to apply test results	6
f. Double check	6
Problem 2	7
a. Updating my data	7
b. Creating captions	10
Problem 3	10
a.	10
b.	11
c.	12
d.	12
e.	13
4.	13
a. Revisit and summarize	13
b. Visual clarity	14
c. Aesthetic clarity	14
d. Recommendations	15

Set-Up Tasks

```
library(tidyverse) # general use
library(here) # cleaning data frames
library(janitor) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # calculating the data
# reading in `temp-kelp` csv file
temp_kelp <- read.csv(
  here("data", "temp-kelp.csv")
)
# reading in `personal_data_wk8` csv file
my_data <- read.csv(
  here("data", "my_es_data.csv")
)
# reading in `bike_to_class` csv file
bike_to_class <- read.csv(
  here("data", "bike_to_class.csv")
)
```

Problem 1

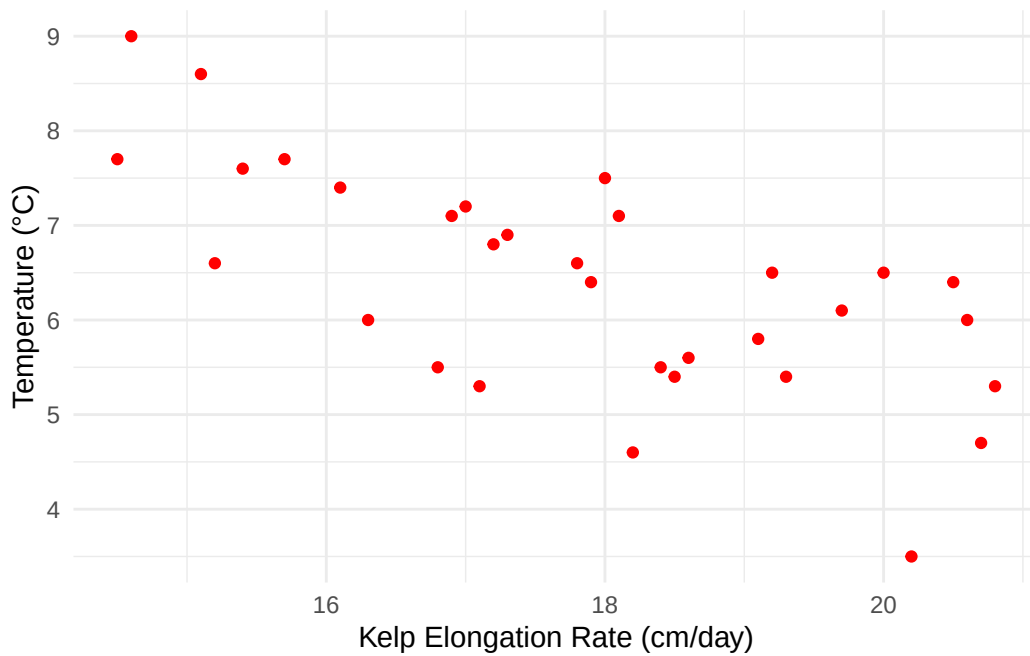
a. Appropriate test

To determine the strength of the relationship between these variables, a Pearson's or Spearman rank correlation can be used. Pearson's correlations measure the direction of a linear relationship with continuous variables, while a Spearman rank correlation measures monotonic relationships between any variables (can be ordinal or continuous).

b. Visualization

```
# base layer: ggplot using `temp_kelp`
ggplot(data = temp_kelp,
  # x-axis: temperature in Celsius
  # y-axis: kelp elongation rate (cm/day)
  mapping = aes(x = temp_c,
    y = kelp_elong)) +
# first layer: geom point to represent individual observations
```

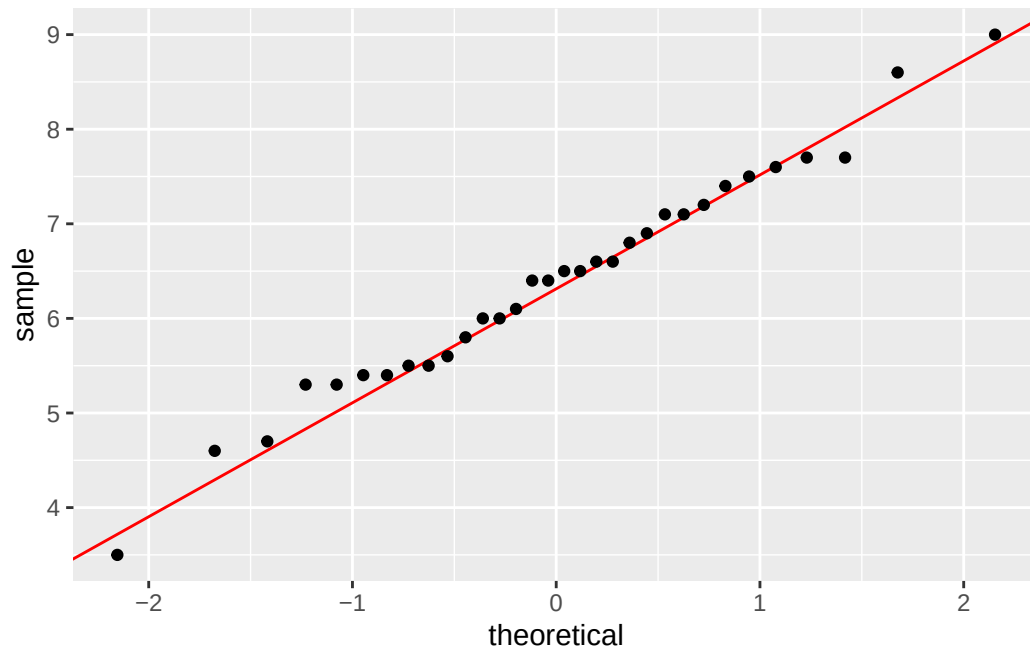
```
# changing color from default to `red`
geom_point(color = "red") +
# relabeling x- and y-axes, adding a title
labs(x = "Kelp Elongation Rate (cm/day)",
      y = "Temperature (\u00B0C)") +
# changing theme
theme_minimal()
```



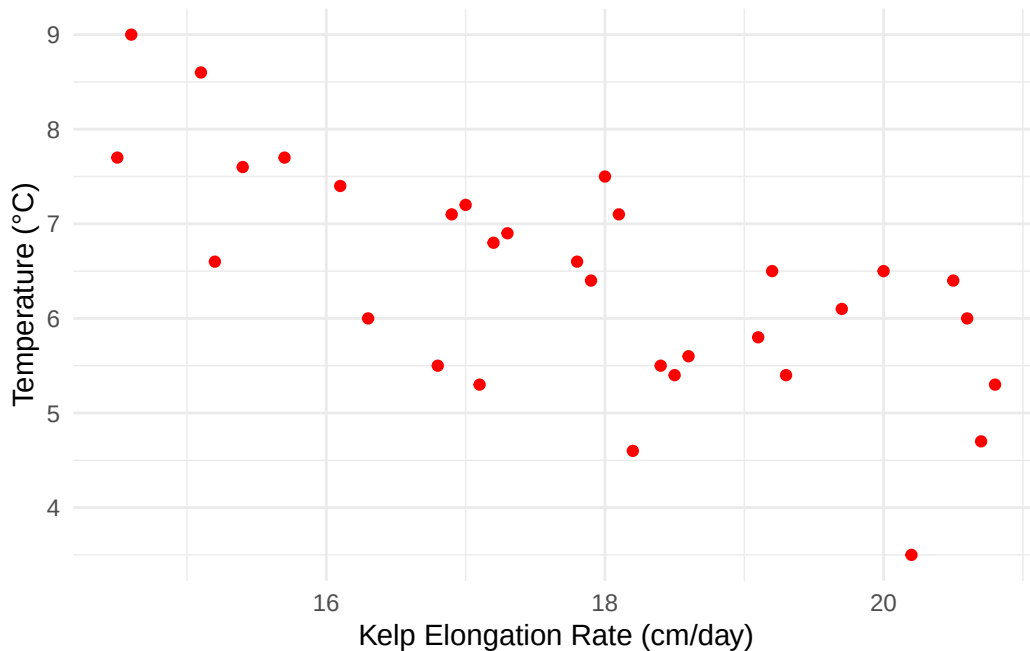
c. Assumption checks

Check assumptions

```
# base layer: ggplot
ggplot(data = temp_kelp, # starting data frame
       aes(sample = kelp_elong)) + # y-axis for QQ plot
# first layer: QQ reference line
geom_qq_line(color = "red") + # adding a color
# second layer: QQ plot
geom_qq()
```



```
# base layer: ggplot using `temp_kelp`
ggplot(data = temp_kelp,
  # x-axis: temperature in Celsius
  # y-axis: kelp elongation rate (cm/day)
  mapping = aes(x = temp_c,
    y = kelp_elong)) +
# first layer: geom point to represent individual observations
# changing color from default to `red`
geom_point(color = "red") +
# relabeling x- and y-axes, adding a title
labs(x = "Kelp Elongation Rate (cm/day)",
  y = "Temperature (\u00B0C)") +
# changing theme
theme_minimal()
```



I checked the assumptions of linearity using a scatter plot and for normality using a Q-Q plot. Given both plots, I would say that this relationship is close enough to be considered normal, considering the minor deviations from the theoretical distribution, there is a mostly linear relationship shown by the general downward pattern of the scatterplot and assumed normality.

Run test

```
# running test to determine strength of correlation
cor.test(temp_kelp$temp_c, temp_kelp$kelp_elong, method = "pearson") # pearson's method
```

Pearson's product-moment correlation

```
data: temp_kelp$temp_c and temp_kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
cor
-0.6877489
```

d. Communication results

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's correlation because the variables were continuous, mostly linear, and normally distributed, enabling me to use this measurement.

We found a moderate negative relationship between temperature and giant kelp frond elongation rate (Pearson's $r = -0.68$, $t(30) = -5.2$, $p < 0.01$, $\alpha = 0.05$), where temperature significantly predicted the decrease of kelp elongation rate.

e. How to apply test results

Given the results of the Pearson's correlation test performed, there is generally a negative relationship between temperature and kelp elongation rate. If the team desires to maximize kelp frond elongation, maintaining a water temperature around 14°C would maintain a higher elongation rate overall, meaning colder ocean environments are more productive. In order to globally conserve kelp species and promote ideal growth conditions, narrowing the team's focus to minimizing climate effects that exacerbate climate temperature increases would aid species propagation for generations to come.

f. Double check

```
# running test to determine strength of correlation
cor.test(temp_kelp$temp_c, temp_kelp$kelp_elong, method = "spearman") # spearman method
```

Spearman's rank correlation rho

```
data: temp_kelp$temp_c and temp_kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

Using a Pearson's versus a Spearman correlation would have made no difference, as in both tests there is a moderate negative relationship temperature and kelp elongation rate and the null hypothesis would be rejected (Spearman $\rho = -0.68$, $S = 9216.1$, $p < 0.01$, $\alpha = 0.05$). While the outputs mean the same thing, the test components differ since a Pearson's correlation test outputs correlation coefficient (r), a one-sample t-test comparing r to 0, and a 95% confidence

interval, while a Spearman correlation test outputs the correlation coefficient (ρ), the sum of all squared rank differences (S), and no confidence intervals. Additionally, within the data, there are tied ranks which leads the p-value computed to be approximate instead of exact for a Spearman correlation.

Problem 2

a. Updating my data

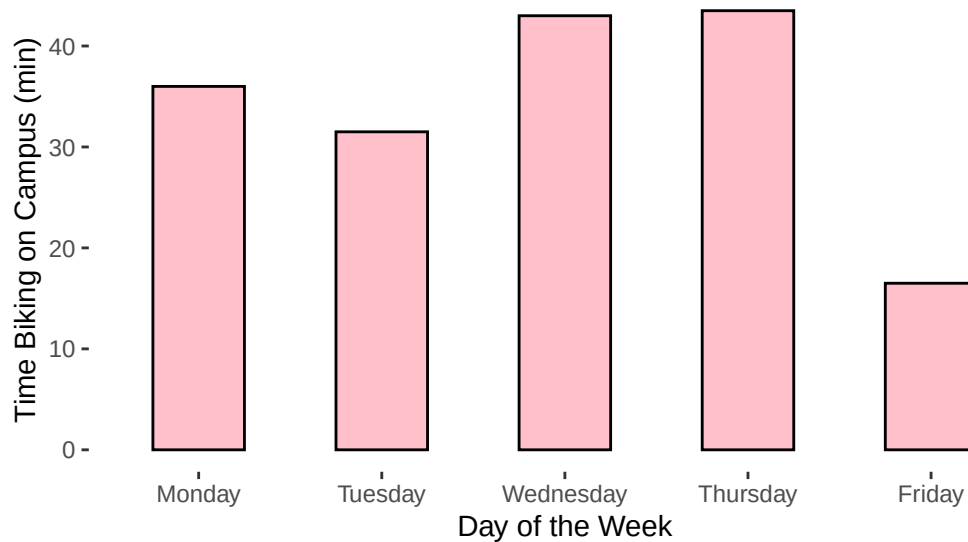
```
# creating new data frame using `my_data`
my_hist <- my_data |>
  # cleaning column names so that only lower case letters and underscores are used
  clean_names()
# reordering levels according to the day of the week in chronological order
my_hist$days <- factor(my_hist$days,
  levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday"))

# creating new data frame using `class_num`
class_num <- bike_to_class |>
  # cleaning column names so that only lower case letters and underscores are used
  clean_names()
```

```
#base layer: ggplot
ggplot(data = my_hist,
  # x-axis should be the day of the week and y-axis should be the time spent biking
  # filling column color to differentiate days
  mapping = aes(x = days,
    y = time_spent_biking_min,
    fill = days)) +
  # first layer: bar chart with height proportional to the values of the data
  geom_col(width = 0.5,
    # color border line of bar black
    color = "black",
    # fill bin color pink
    fill = "pink") +
  # relabeling x- and y-axes, adding title and subtitle
  labs(title = "Comparison of total time spent biking throughout the week",
    subtitle = "May 22, 2026",
    x = "Day of the Week",
    y = "Time Biking on Campus (min)") +
```

```
# changing theme of histogram
theme(panel.background = element_blank()) +
# removing legend
theme(legend.position = "none")
```

Comparison of total time spent biking throughout the week
May 22, 2026

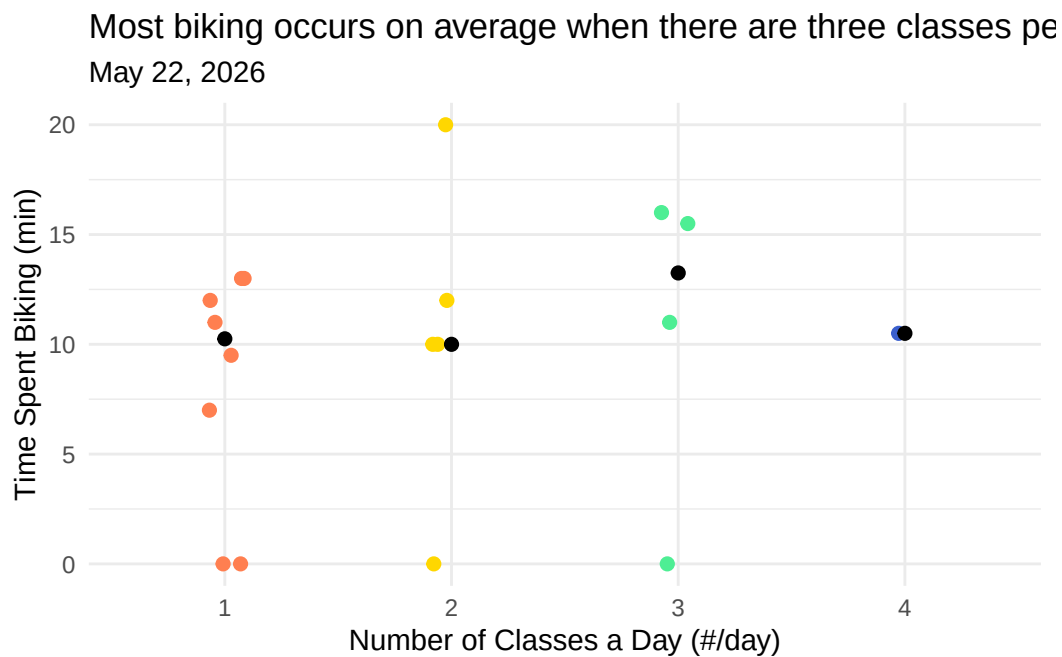


```
# base layer: ggplot
ggplot(data = class_num,
# x-axis: number of classes per day
# y-axis: time spent biking (min)
# coloring by number of classes per day
mapping = aes(x = factor(number_of_classes_per_day),
y = time_spent_biking_min,
color = factor(number_of_classes_per_day))) +
# first layer: jitter plot
# coloring data points by number of classes in custom coloration
geom_jitter(height = 0,
width = 0.1,
size = 2,
aes(color = factor(number_of_classes_per_day))) +
scale_color_manual(values = c("1" = "coral",
"2" = "gold",
"3" = "seagreen2",
```

```

"4" = "royalblue3")) +
# displaying medians for each level along the x-axis
# changing color and size
stat_summary(
  fun = median,
  geom = "point",
  size = 2,
  color = "black"
) +
# relabeling x- and y-axes, adding a title and subtitle
labs(title = "Most biking occurs on average when there are three classes per day",
      subtitle = "May 22, 2026",
      x = "Number of Classes a Day (#/day)",
      y = "Time Spent Biking (min)") +
# changing theme to remove any visual clutter
theme_minimal() +
# removing legend
theme(legend.position = "none")

```



b. Creating captions

Figure 1. The amount of time biking (min) differs across days of the week. The pink bins of the bar graph represent the cumulative number of minutes spent biking on campus during that day of the week over four weeks.

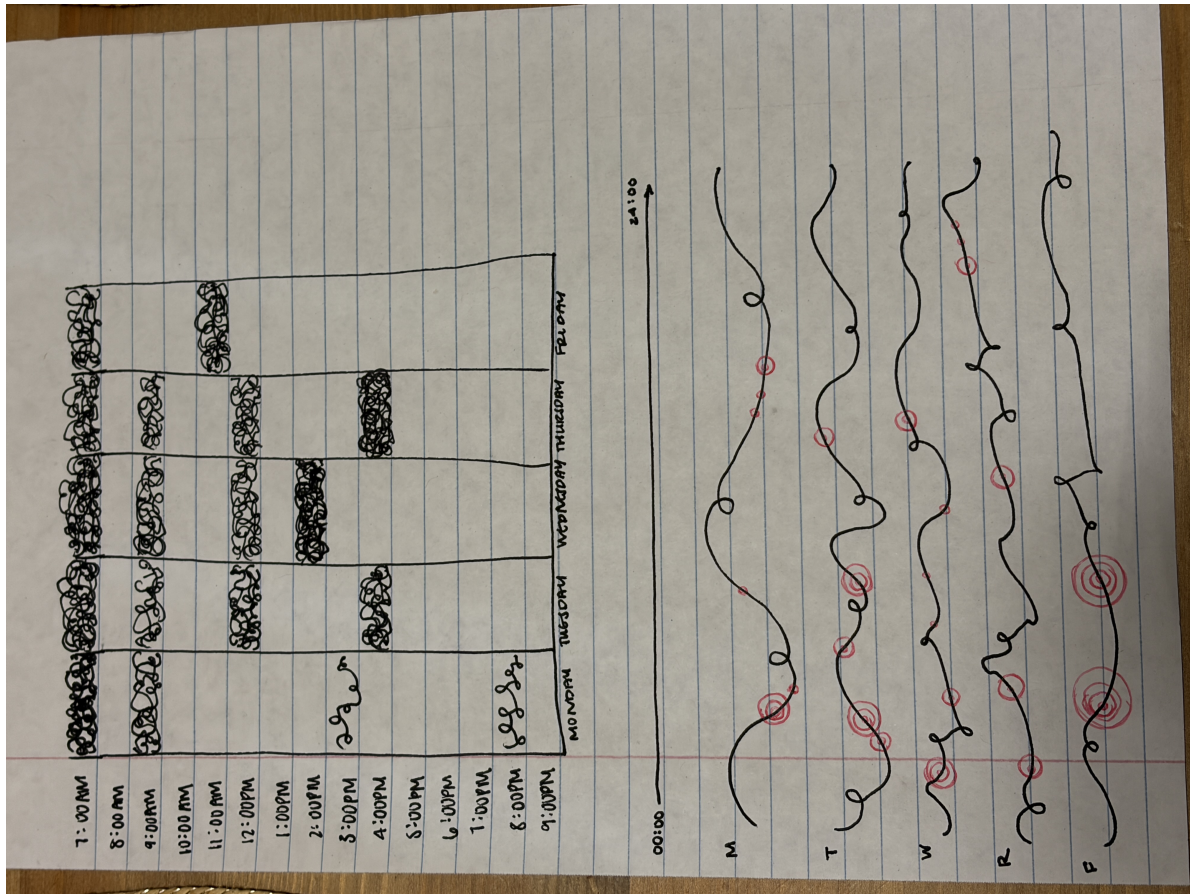
Figure 2. I spend the most time biking (min) on average when there are three classes in a day. Each colored point, aside from black, represents measurements of time spent biking (min) across a different number of classes (1-4). Colors differentiate the number of classes had during a given day (coral: one class, yellow = two classes, green = three classes, blue = 4 classes). Black points represent the median amount of time biking on campus given a certain number of classes.

Problem 3

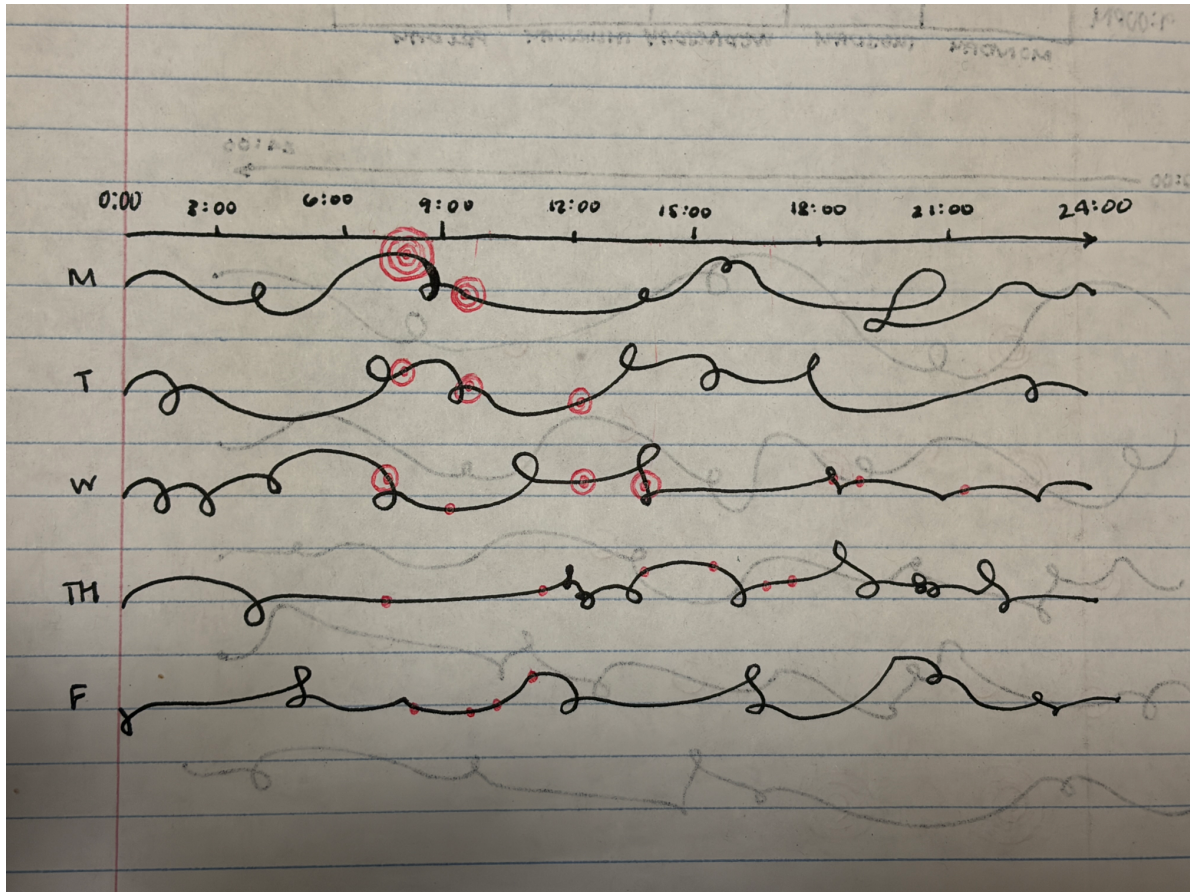
a.

As my data is about time spent biking on campus, I could focus on the stress and anxiety of getting to class on time. The quicker I am biking to my next destination is likely reflected by a faster bike ride, and so the use of darker and more brooding colors could reflect this. Additionally, the mental mania and obsession associated with academics, timeliness, and fitting everything in could be represented with purposefully disorienting, and maybe even lines of varying widths with jagged angles. The idea is to create a visual affective piece that evokes a feeling of stress and perpetual belatedness.

b.



c.



d.

My piece shows the time of day over the weekdays in which I am biking to class, with each additional circle representing my repetition of biking. Each week was represented by a “line in turmoil,” where the stress of the week and biking faster and faster from place to place was catching up to me. The colors of black and red are also meant to work with color theory such that the viewer is put at unease.

I was influenced to draw in this style by Stefanie Posavec and Giorgia Lupi’s “Dear Data project,” where weeks were shown and time was recorded in wacky yet meaningful ways. I also found freehand drawing with no real pattern to be perfect for the “graph lines,” as the stress a student feels comes from all angles at times, and the mind is constantly at unrest.

The form of my work in this rough draft is purely pen work, but in a final draft I would like to add water color pens and a legend.

I was originally quite uninspired because biking doesn't seem to have much emotion behind it. However, as I was thinking to my current state of stress, my need to bike faster and faster from place to place, and the lack of "enough time" during the day, I found there to be a large mental health aspect inadvertently related to biking on school campuses. When creating my work, I used the lines on my paper to create what I call "type-B precision," where each week begins at the beginning of the red line, but goes berzerk following that, and I also used multiple aspects of my personal data, implementing it as Posavec and Lupi do so well.

e.

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4.

a. Revisit and summarize

The paper that I selected in homework 1 uses a Mann-Whitney U test as well as Kruskal Wallis test. However, I generally have been focusing on the Mann-Whitney U component of this paper. In this case, a Mann-Whitney U test addresses one of the main questions of the possible effects of predation by finfish on sea urchin community structures, indicated by a protected (isolated from fisherman and with abundant finfish) or unprotected (not isolated from fisherman and with lower finfish abundance) reef, rather than focusing on changes in urchin population density variance due to normalization.

Table 4. Total sea urchin densities and survival values (see methods) for protected and unprotected reefs. The significance level of a Mann-Whitney *U* test of difference between protected and unprotected reefs is provided

	Protected	Unprotected	Mann-Whitney <i>U</i> test
Sites (<i>n</i> =)	3	11	
Transects (<i>n</i> =)	24	96	
Total sea urchin density (#/10 m ²)	0.45 ± 0.12	80.70 ± 7.30	<i>p</i> < 0.001
Survival (days)	0.40 ± 0.08	1.68 ± 0.07	<i>p</i> < 0.001

b. Visual clarity

Viewing the table I inserted above, the authors clearly visually represented their statistics by placing the response variables in the rows (total sea urchin density and survival) and the predictor variable and its orders in the columns (reef protection - protected vs. unprotected), as well as whether the p-value is significant or not. For each of these outcomes, a p-value will indicate whether there is a significant difference between fisherman-protected and unprotected reefs on fish shown in a clear manner. Additionally these outcomes each come with a SE, which is some number above or below the calculation numbers.

c. Aesthetic clarity

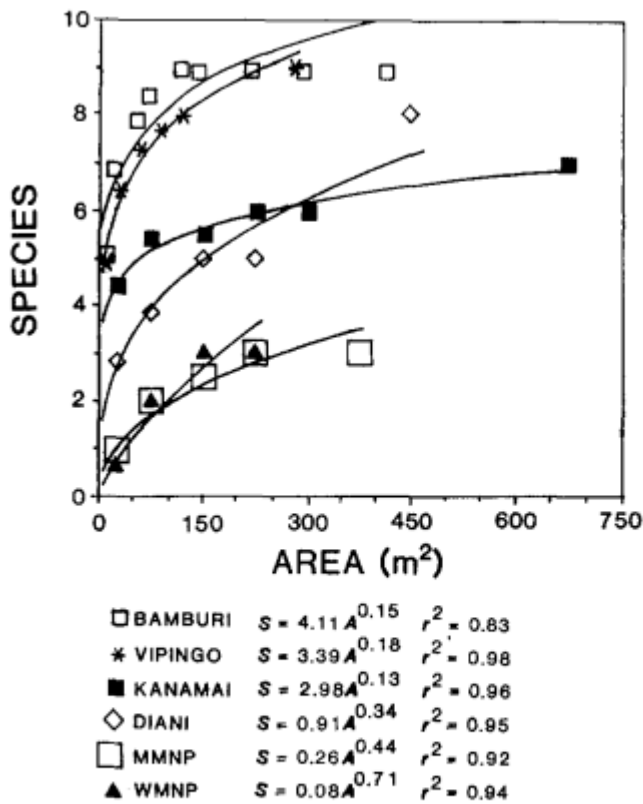


Fig. 2. Species-area curves and best-fit equations for sea urchins inhabiting reef lagoons at the six reefs sampled. MMNP and WMNP = Malindi and Watamu Marine National Parks

Within this figure, meant to show the differing groups of urchin species depending on how large the area of observation is, there is generally a low data:ink relationship. This is likely due to the overlapping slope lines shown, and even with differing symbols for each reef lagoon,

it is still visually busy and a bit confusing. The most confusing aspects of this graph are the outliers which overlap and cross over other lines, furthering confusion of the viewer.

d. Recommendations

To make the figure 2 table more clear and visually appealing, there is the potential to split the graph into multiple single species-area curves to reduce overlap confusion, omit the use of best-fit lines and connect the individual points, or color each line and the corresponding points. Splitting up the graphs into six smaller ones is definitely not the most efficient and would take up more space, but would allow the comparison of individual points to a greater extent. Connecting the points rather than using a line of best fit would change each shape overall, but could drastically increase precision and clarity of data. Lastly, and my preferred solution, would be to pick individual colors for each reef region, not changing the data or methods themselves, but bettering the ability to pick out points to match with said region.